
MODEL SELECTION AND VALIDATION REPORT (remove_4_biomass model)

OBSERVATIONS:1716

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_4_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	6.30	0.46
remove_4_tric	0.31	0.03
remove_4_teve	-0.28	0.02
sbt_mean	0.16	0.09
sbt_sd	-0.03	0.04
sst_sd	0.05	0.04
log_chla_mean	0.03	0.03
depth	0.06	0.04
log_sum_fishing	0.08	0.03

Dispersion parameter: 0.82

Spatiotemporal AR1 correlation (rho): 0.97

Matérn range: 423.97

Spatiotemporal marginal AR1 SD: 1.09

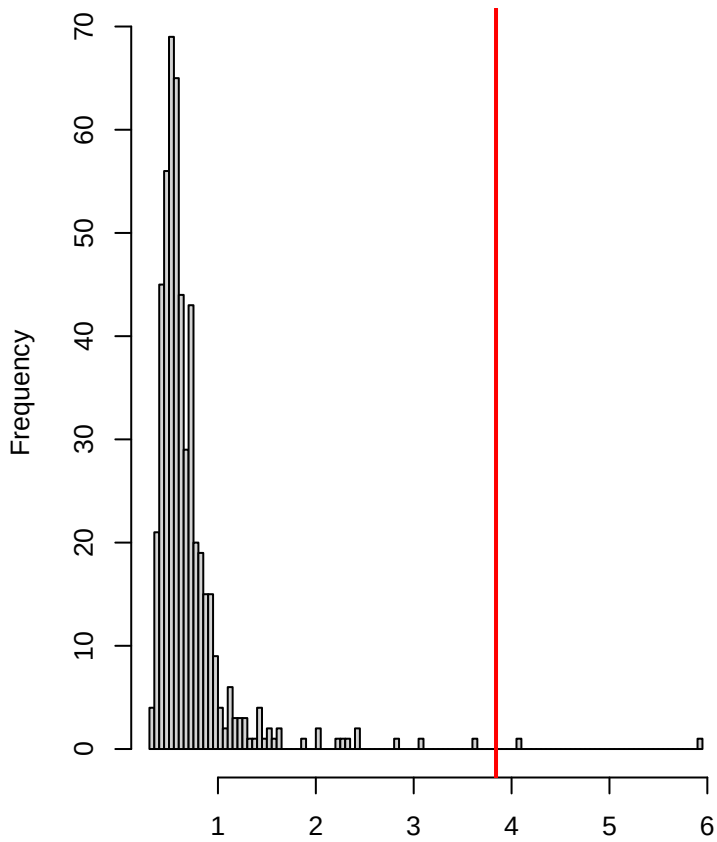
REML criterion at convergence: 12153.424

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

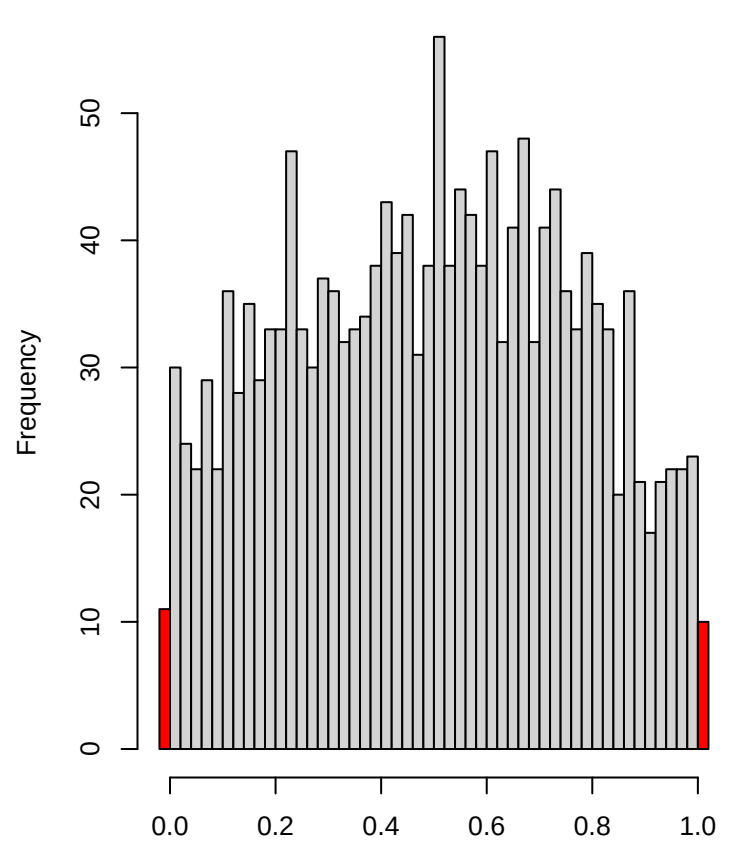
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	24332.85	3.227403
spatiotemporal_random_fields_ar1_with_anisotropy	15	24336.07	138.361303
spatial_random_fields	12	24474.44	2.900832
spatial_random_fields_with_anisotropy	14	24477.34	45.230877
spatiotemporal_random_fields_iid_with_anisotropy	14	24522.57	2.292799
spatiotemporal_random_fields_iid	12	24524.86	226.858219
error_only	10	24751.72	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



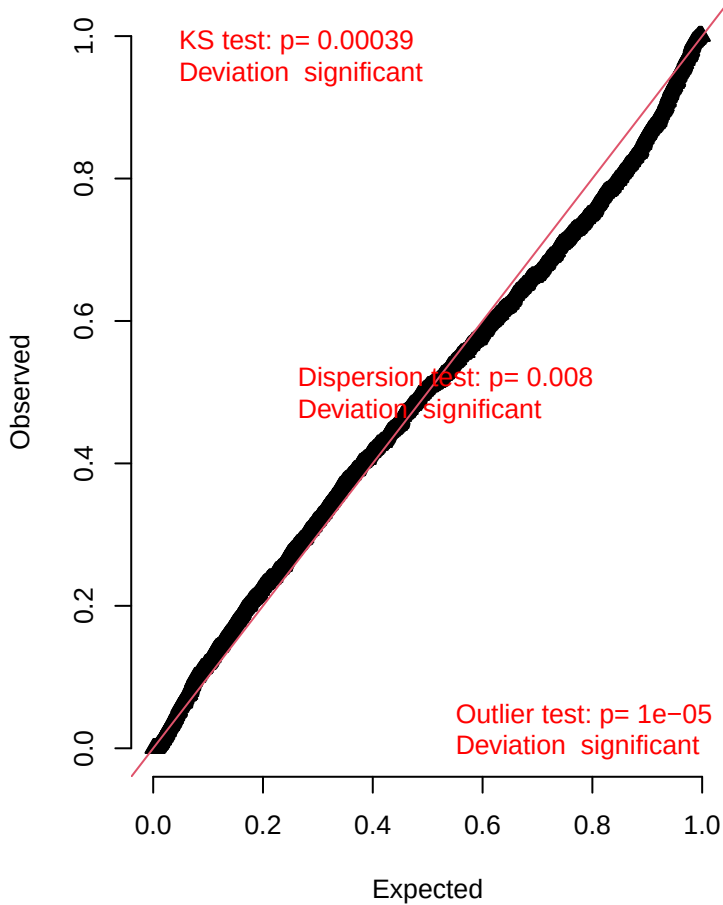
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

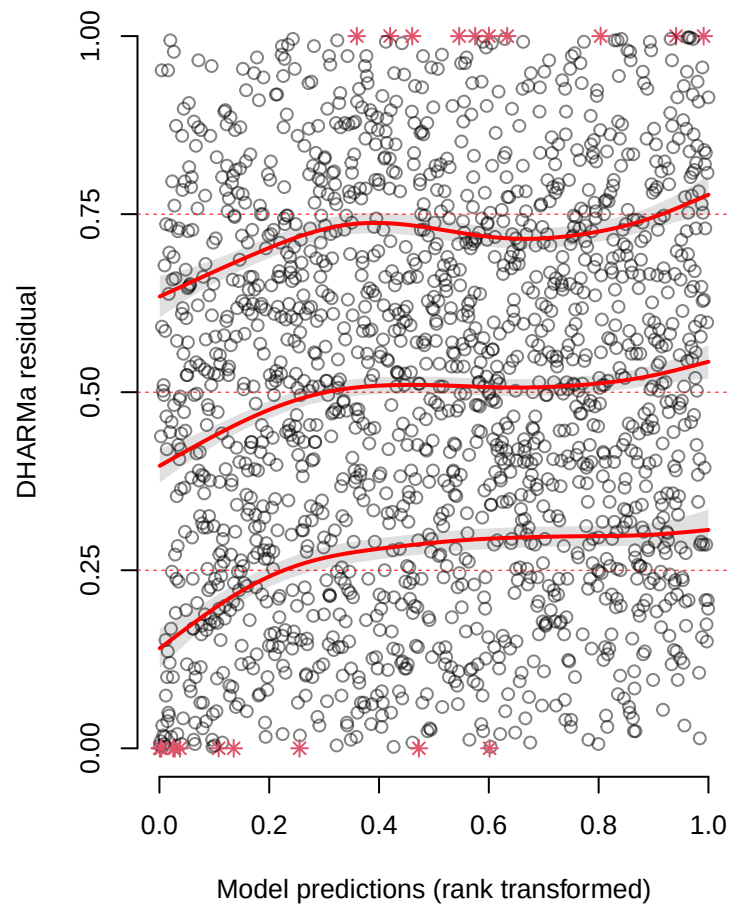


Residuals (outliers are marked red)

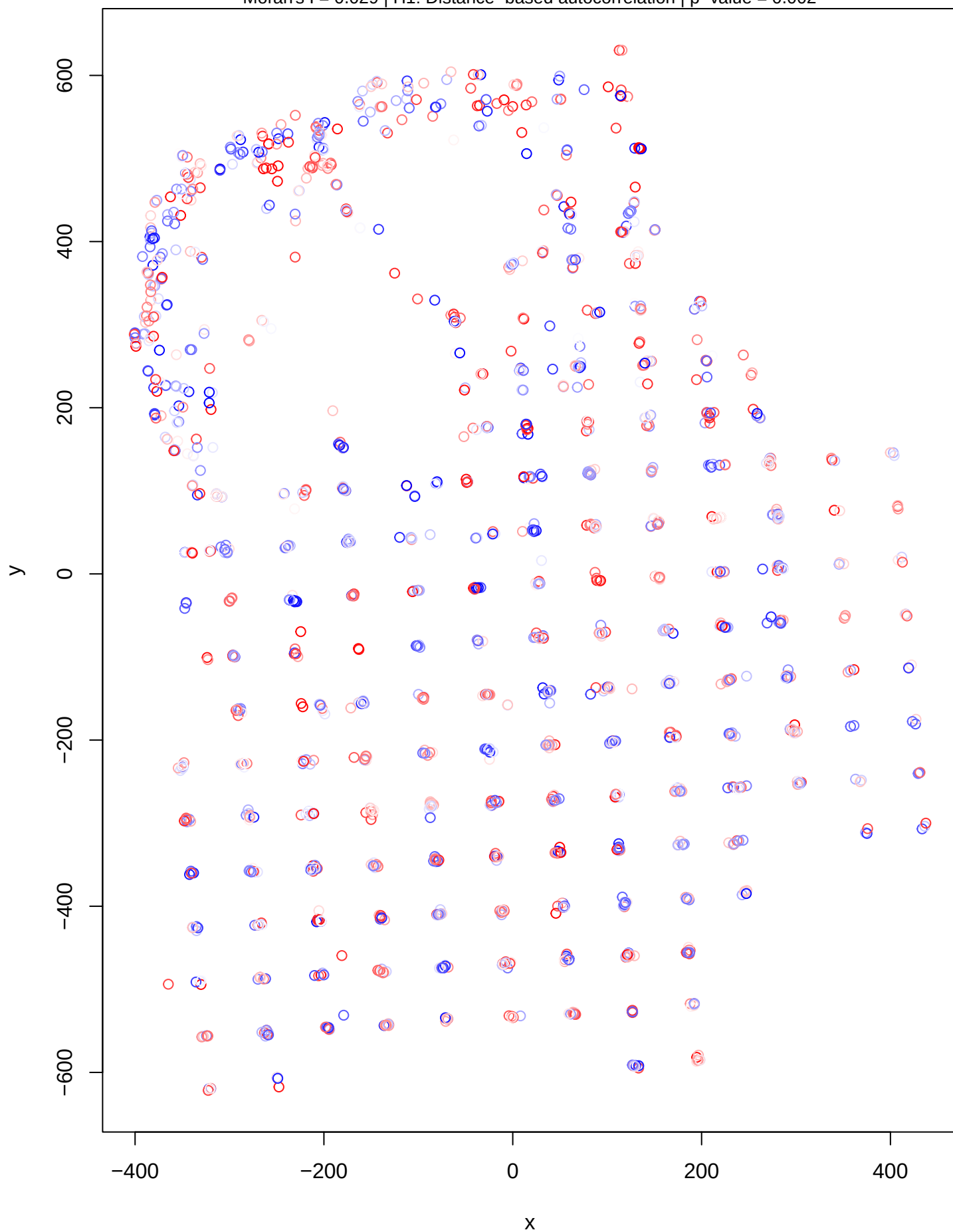
QQ plot residuals



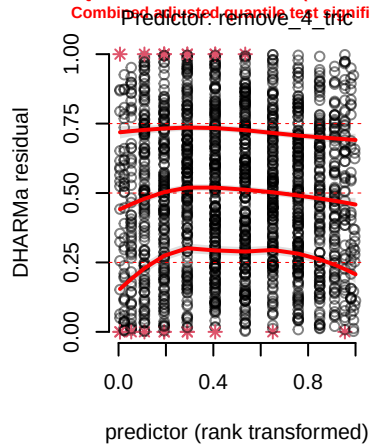
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



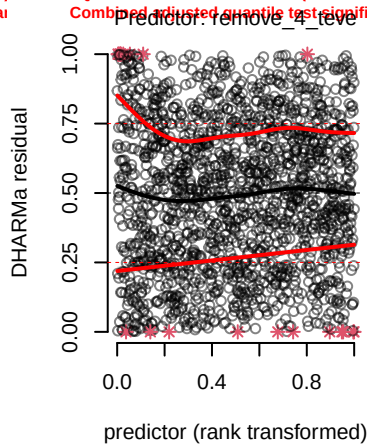
DHARMA Moran's I test for distance-based autocorrelation
Moran's I = 0.029 | H1: Distance-based autocorrelation | p-value = 0.002



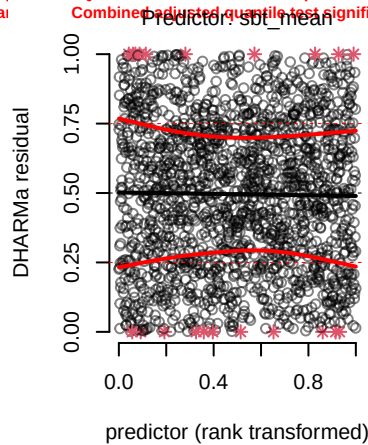
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



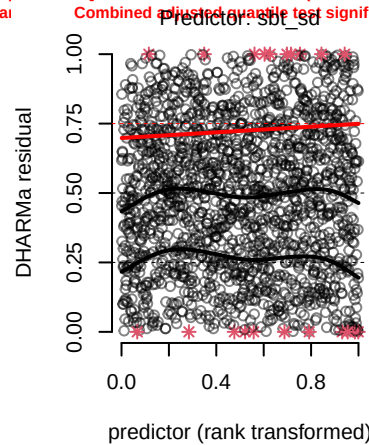
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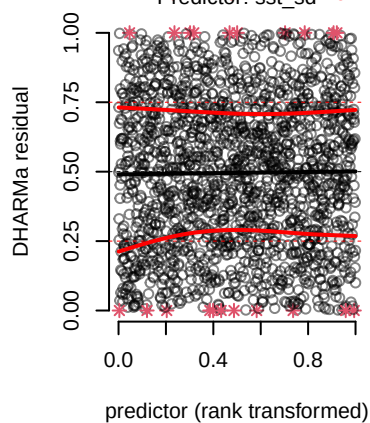
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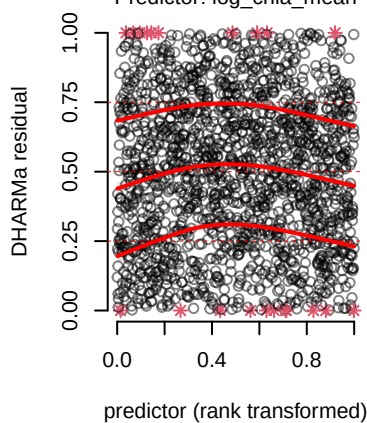
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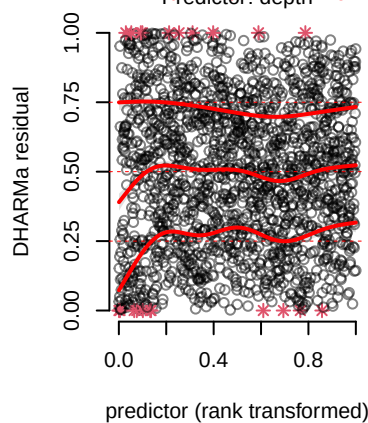
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